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Lumpy Skin Disease in Cattle and Buffalo in Asian Countries: A Systematic Review and Meta-Analysis

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ABSTRACT

Background: Lumpy skin disease (LSD) is an acute viral transboundary disease of cattle and buffalo with significant economic importance caused by the LSD virus in the *Poxviridae* family.

Objective: This study aimed to determine the regional prevalence of LSD in cattle and buffalo in Asian countries using a systematic approach.

Methods: A systematic search for articles on the prevalence of LSD in cattle and buffalo was performed via PubMed, Web of Science, Scopus, Google Scholar, and article reference lists published between 1 January, 2012 and 28 February, 2025. Articles were selected according to the established inclusion and exclusion criteria. The pooled prevalence was estimated through a random effects meta-analysis model because significant heterogeneity existed among the studies.

Results: The overall pooled estimated prevalence of LSD in Asia was 25.2% (95% CI: 19.95–30.88%), with significant variations across species, countries, and diagnostic methods. Cattle had the highest pooled prevalence at 26.2% (95% CI: 20.73–31.96%), whereas buffalo reported much lower rates at 7.7% (95% CI: 3.60–13.00%), and epidemic cases presented the highest prevalence at 41.4% (95% CI: 0–98.1%). Temporal analysis revealed that the peak prevalence in 2016 was 74% (95% CI: 68.56–79.06%), and the lowest estimate in 2015 was 16.3% (95% CI: 3.25–36.47%).

Conclusion: This study reveals significant regional and species-based variations in LSD prevalence across Asia, with diagnostic methods and temporal trends influencing the results. These findings underscore the necessity for targeted surveillance and control strategies to address high-prevalence areas.

Abbreviations: CI, Confidence interval; DNA, Deoxyribonucleic Acid; ELISA, Enzyme-linked immunosorbent assay; LSD, Lumpy skin disease; LSDV, Lumpy skin disease virus; PCR, Polymerase chain reaction; PRISMA, Preferred Reporting Items for Systematic Reviews and Meta-analyses; RT-PCR, Real-Time polymerase chain reaction; SE, Standard error; WOA, World Organization for Animal Health.

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1 | Introduction

Lumpy skin disease (LSD) is an acute or subacute viral disease in cattle that can cause fever, depression, skin nodules (undergo necrosis), reduced milk production (more than 50%), and low mortality (1–3%) but high morbidity (5–45%) and is influenced by breed, host immunity, and insect vectors (Coetzer and Tuppurainen 2004; Tuppurainen and Oura 2012; Mafirakureva et al. 2017; Abdullatif and Allawe 2021). LSD is also referred to as ‘Neethling virus disease’, ‘exanthema nodularis bovis’, ‘pseudourticaria’, and ‘knopvelsiekte’, with LSD being the most widely recognised and commonly used term (Al-Salihi 2014; Tuppurainen et al. 2017). LSD is a transboundary vector-borne disease primarily transmitted through blood-feeding arthropods, including mosquitoes and *Stomoxys* spp., as well as via contaminated feed, water, and bodily secretions (Buller et al. 2005; Sprygin et al. 2019). All ages and breeds of cattle and water buffalo are susceptible to LSD; however, the disease is most frequently reported and more severe in young, underweight, immunocompromised cattle and those in peak lactation (Weiss 1968; Khan et al. 2021). A warm, humid climate; water availability; new animal introductions; the rainy season; and cattle grazing in wet areas are key risk factors for LSD virus (LSDV) outbreaks (Davies 1982; Dawoud et al. 2019). Owing to its rapid transboundary spread, WOAHP classifies LSD as a notifiable disease with severe impacts on the international livestock trade, but its environmental persistence and multiple transmission routes pose significant challenges for global control and prevention (Swiswa et al. 2017; Tuppurainen et al. 2018; Gupta et al. 2020; Wilhelm and Ward 2023). LSD causes poor reproduction, degraded hide quality, reduced milk yield, emaciation, and economic losses for farmers, with its emergence in South Asia raising concerns over production losses, reduced draught power, lower feed intake, trade restrictions, and prolonged recovery (Kumar et al. 2018; Gupta et al. 2020).

Since its emergence in Zambia, Africa, in 1929, LSD has spread globally for more than 90 years, becoming endemic across Africa and rapidly reaching previously disease-free countries. (Macdonald 1931; Li et al. 2023). LSD was first reported in the Middle East in 1988 in Egypt and in 2005 in Bahrain, remaining confined to the region (Western Asia) until 2018. (Stram et al. 2008; Roche et al. 2021). LSD was first reported in South Asia in 2019 and affects China, Bangladesh, India, and Nepal (Hasib et al. 2021; Roche et al. 2021). India’s first outbreak occurred in Odisha before it spread nationwide (Authority et al. 2020). By 2020, the disease had spread to Nepal, Sri Lanka, Bhutan, Bangladesh, Vietnam, and Southeast China (Acharya and Subedi 2020; Roche et al. 2021; Tran et al. 2021). Bangladesh emerged as the first hotspot, with the earliest case recorded on 14 July 2019 (Pal and Gutama 2023). In 2021, LSD spread to Malaysia, Thailand, and Cambodia, leading to widespread outbreaks across Southeast Asia and India (Roche et al. 2021; Li et al. 2023).

This disease has emerged as a devastating threat to large domesticated ruminants in Asia, Europe, and the Middle East (Authority et al. 2020). Moreover, the current LSD epidemic presents a major threat to Asia for three main reasons. First, disease transmission dynamics, including variations across ecological and climatic conditions and the exact role of mechanical vectors, remain poorly understood. Second, as an emerging transboundary dis-

ease, awareness among veterinary professionals and farmers is limited, leading to underreporting and misdiagnosis, which hinders effective prevention efforts. Third, the lack of effective and widely available vaccines further complicates control measures, increasing the risk of continued outbreaks. Goat and sheep pox vaccines exhibit efficacy but are associated with potential administration risks. However, the commercially available live attenuated LSD vaccine is banned in several countries, including the Russian Federation (Brenner et al. 2009; Beard 2016), due to its use having been associated with outbreaks caused by vaccine-like Neethling strains, exemplified by the 2017 incident in cattle (Kononov et al. 2019). Nevertheless, the extensive administration of live attenuated vaccines during the 2015–2017 Balkans outbreak effectively contributed to controlling the epidemic (Tuppurainen et al. 2020).

Despite the significant economic impact of LSD, research on this highly devastating arthropod-borne disease remains limited in South Asian and East Asian countries (Kayesh et al. 2020; Sudhakar et al. 2020). Therefore, the absence of a government control plan or contingency for LSD, coupled with limited knowledge of its status and outbreak trends in the region, highlights a critical gap in disease management. Consequently, the repeated outbreaks and global resurgence of LSD emphasise the necessity of reassessing its disease dynamics, viral transmission pathways, and modernised preventive and control measures. Based on these insights, this study aims to systematically review and conduct a meta-analysis of the prevalence of lumpy skin disease in cattle and buffalo from 2012 to 2025, with a particular emphasis on data from Asian countries to better understand regional disease patterns and inform control strategies.

2 | Materials and Methods

The study adhered to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines for conducting a systematic review and meta-analysis. To ensure the inclusion of all relevant information and uphold methodological standards, the PRISMA 2009 checklist was followed (Supplementary Material S1).

2.1 | Search Strategy

A modified systematic search method was employed to identify all published research concerning the prevalence of LSD in cattle and buffalo. The following electronic databases—Google Scholar, PubMed, Springer Nature Link, and Web of Science—were used to retrieve articles published from January 2010 to February 2025. The investigation was carried out on 19 February 2025. The following keywords were used to search the literature: (Prevalence OR Epidemiology OR Seroprevalence OR Incidence OR Detection OR Occurrence OR Identification OR Molecular characterisation) AND (LSD OR Lumpy skin disease) AND (Cattle OR Buffalo) AND (Asia). A language restriction to English was applied, with search terms and keywords modified on the basis of the syntax requirements of the four electronic databases (Moher et al. 2009).

The inclusion process adhered to the PRISMA guidelines. A PRISMA checklist was utilised to ensure that the systematic review and meta-analysis incorporated all essential information.

2.2 | Selection of Studies

Articles were selected for meta-analysis on the basis of the following criteria: full-text articles published in English between January 2012 and February 2025; studies conducted in Asian countries; studies involving cattle or buffalo or both; and studies reporting the prevalence of LSD in animals. The criteria for excluding articles were species other than cattle and buffalo, the prevalence was not presented, flocking with a history of vaccination, comparisons of methods, experimental trials, and articles in languages other than English.

2.3 | Eligibility Criteria

The search process was conducted independently by five authors (MJA, MAR, PB, MJC, and MAR). The eligibility criteria for selecting articles were as follows: (1) research articles carried out in Asia and previously published in peer-reviewed journals; (2) studies reporting the prevalence of LSD, epidemic, endemic, or outbreak situations; (3) articles providing comprehensive information; (4) studies employing serological ELISA (unvaccinated), PCR, real-time PCR (qPCR), necropsy, and clinical signs diagnosis; (5) articles presenting data on total sample size and detecting positive cases; (6) research articles available in the English language; and (7) studies published from 2012 to 2025. Articles that failed to match the criteria mentioned above were excluded.

2.4 | Quality of the Studies

A quality assessment checklist was used to evaluate the reporting quality and selection bias of the studies included in this meta-analysis (Ahaduzzaman 2019). The checklist contains seven parameters, scoring responses as 1 for 'yes' and 0 for 'no', with a final mean score computed for each article. The articles were classified into three categories (3): low quality = 0–3; moderate quality = 4–5; and high quality = 6–7 (Supplementary Materials S2 and S3).

2.5 | Data Extraction

Data, including authors, year of publication, country (Asia), animal species, types of diagnostic tests, total sample size, and total positive samples, were retrieved from each eligible study into a spreadsheet. A total of 166,282 animals (cattle: 165,955 and buffalo: 327) from diverse geographical regions were analysed (Table 1).

2.6 | Data Analysis

The data that were retrieved were then entered into an Excel spreadsheet after being transcribed. (Elliott et al. 2006). By using

a random effects model meta-analysis, the pooled prevalence of LSD at 95% CI was determined (DerSimonian and Laird 1986; Abebaw 2024). Heterogeneity is classified as low, moderate, or high at I^2 values of 25%, 50%, and 75%, respectively, with 0% indicating its absence (Higgins and Thompson 2002). A random-effects meta-analysis was chosen for summary statistics because of the substantial heterogeneity across studies. Subgroup analysis was also carried out by country, diagnosis, species, year, and disease occurrence. A random-effects meta-regression was performed to analyse the prevalence of LSD in cattle and buffalo over time. Egger's test and a funnel plot were used to assess publication bias, along with tests to identify sources of asymmetry (funnel plot) and small study effects (Egger et al. 1997). A p -value < 0.05 on the Egger test was considered indicative of statistically significant publication bias. A sensitivity analysis was conducted by systematically removing one study at a time to assess the stability and reliability of the pooled prevalence estimate. Meta-analysis of the data was conducted using Stata version 18.0 (College Station, TX, USA). Different R packages (version 4.4.2), including 'spdep' and 'ggplot' for maps (Wickham 2011; Bivand et al. 2017), were used for this study. This systematic review and meta-analysis were based on PRISMA Statement (Moher et al. 2009).

3 | Results

3.1 | Study Selection

A systematic search identified 1240 records from electronic databases. After 449 duplicate records and 12 ineligible records were removed, 779 unique studies were screened. Of these, 324 were excluded on the basis of title and abstract evaluation. Full-text retrieval was attempted for 455 studies, but 267 were unavailable. Among the 188 full-text articles assessed for eligibility, 141 were excluded because of the unavailability of the full text ($n = 39$), lack of individual prevalence data ($n = 67$), non-English language ($n = 7$), or other reasons ($n = 28$). Ultimately, 47 studies met the inclusion criteria and were included in the final review and meta-analysis (Figure 1). Thirteen studies from Bangladesh, seven studies from India, two studies from Pakistan, four studies from Iraq, three studies from Nepal, one study from South Korea, two studies from Indonesia, one study from Oman, one study from Afghanistan, one study from Azerbaijan, two studies from Iran, one study from Kazakhstan, one study from Saudi Arabia, one study from Myanmar, two studies from China, two studies from Turkey, one study from Jordan, and two from Thailand.

3.2 | Descriptive Characteristics of the Included Studies

Among the 1240 studies identified through the literature search, 47 studies were included on the basis of the criteria in this systematic review and meta-analysis, covering the prevalence of LSD in 18 Asian countries between 2012 and 2025 (Figure 1). The diagnosis of LSD was primarily based on RT-PCR (6.38%), PCR (31.92%), ELISA (8.51%), and overt clinical signs (clinical examination) (53.19%). Some studies relied on official veterinary surveillance reports, whereas others used farm surveys and outbreak reports to estimate disease prevalence.

TABLE 1 | Summary of metadata of LSD from 2012–2025 published studies in Asia.

Country	Author	Year	Disease occurrence	Sample size	Positive sample	Type of diagnostic test	Species
Bangladesh	(Pory et al. 2021)	2021	Endemic	2762	377	Clinical signs	Cattle
	(Chouhan et al. 2022)	2022	Endemic	1187	403	PCR	Cattle
	(Uddin et al. 2022)	2022	Endemic	8215	2175	Clinical signs	Cattle
	(Hasib et al. 2021)	2021	Outbreak	3327	345	Clinical signs	Cattle
	(Biswas et al. 2024)	2024	Endemic	2511	426	Clinical signs	Cattle
	(Rafe-Ush-Shan et al. 2025)	2025	Outbreak	200	52	PCR	Cattle
	(Islam et al. 2023)	2023	Endemic	690	94	Clinical signs	Cattle
	(Haque et al. 2021)	2021	Endemic	87	28	Clinical signs	Cattle
	(Parvin et al. 2022)	2022	Outbreak	165	86	RT–PCR	Cattle
	(Nobel et al. 2023)	2023	Endemic	634	203	Clinical signs	Cattle
	(Parvin et al. 2025)	2025	Endemic	1161	304	ELISA	Cattle
	(Khalil et al. 2021)	2021	Outbreak	726	139	Clinical signs	Cattle
	(Biswas et al. 2020)	2020	Outbreak	183	109	Clinical signs	Cattle
India	(Sethi et al. 2022)	2022	Outbreak	452	63	PCR	Cattle
	(Smaraki 2022)	2022	Endemic	660	359	ELISA	Cattle
	(Nayakvadi et al. 2022)	2022	Endemic	1471	68	PCR	Cattle
	(Sudhakar et al. 2020)	2020	Outbreak	2539	182	Clinical signs	Cattle
	(Akhter et al. 2025)	2025	Outbreak	1661	643	PCR	Cattle
	(Lakshmi Kavitha et al. 2021)	2021	Outbreak	405	82	PCR	Cattle
Pakistan	(Sudhakar et al. 2025)	2025	Epidemic	177	18	RT–PCR	Buffalo
	(Jabbar et al. 2025)	2025	Endemic	800	290	PCR	Cattle
	(Khan et al. 2024)	2024	Epidemic	385	296	PCR	Cattle
Iraq	(Wajid 2017)	2017	Outbreak	247	22	Clinical signs	Cattle
	(Jarullah 2015)	2015	Endemic	2906	249	Clinical signs	Cattle
	(Aboud and Luaibi 2022)	2022	Endemic	150	8	PCR	Buffalo
	(AL-ETHAFA 2021)	2021	Outbreak	63	17	PCR	Cattle
Nepal	(Pandey et al. 2021)	2021	Endemic	107	57	ELISA	Cattle
	(Dhakal et al. 2024)	2024	Outbreak	1538	431	Clinical signs	Cattle
	(Acharya and Subedi 2020)	2020	Outbreak	13.900	1395	Clinical signs	Cattle
South Korea	(Kim et al. 2024)	2024	Outbreak	3910	1196	ELISA	Cattle
Indonesia	(Hidayat et al. 2025)	2025	Endemic	458	129	Clinical signs	Cattle
	(Murti et al. 2024)	2024	Outbreak	4373	454	Clinical signs	Cattle
Oman	(Body et al. 2012)	2012	Outbreak	201	56	PCR	Cattle
Azerbaijan	(Zeynalova et al. 2016)	2016	Outbreak	269	199	PCR	Cattle
Iran	(Sameea Yousefi et al. 2017)	2017	Outbreak	683	122	Clinical signs	Cattle
	(Sameea Yousefi et al. 2018)	2018	Endemic	234	80	PCR	Cattle
Kazakhstan	(Issimov et al. 2022)	2022	Endemic	8607	881	Clinical signs	Cattle
Saudi Arabia	(Kasem et al. 2018)	2018	Outbreak	64.109	3852	Clinical signs	Cattle
China	(Lu et al. 2021)	2021	Outbreak	801	156	Clinical signs	Cattle
	(Song et al. 2024)	2024	Outbreak	41	12	PCR	Cattle
Turkey	(Şevik and Doğan 2017)	2017	Endemic	611	440	RT–PCR	Cattle
	(Mat et al. 2021)	2021	Outbreak	28.434	1282	Clinical signs	Cattle

(Continues)

TABLE 1 | (Continued)

Country	Author	Year	Disease occurrence	Sample size	Positive sample	Type of diagnostic test	Species
Jordan	(Abutarbush et al. 2015)	2015	Endemic	624	162	Clinical signs	Cattle
Thailand	(Arjkumpa et al. 2024)	2021	Outbreak	1274	516	Clinical signs	Cattle
	(Suwankitwat et al. 2022)	2022	Outbreak	859	426	PCR	Cattle
Myanmar	(Maw et al. 2022)	2022	Outbreak	180	13	Clinical signs	Cattle
Afghanistan	(Sangary et al. 2023)	2023	Endemic	1305	291	Clinical signs	Cattle

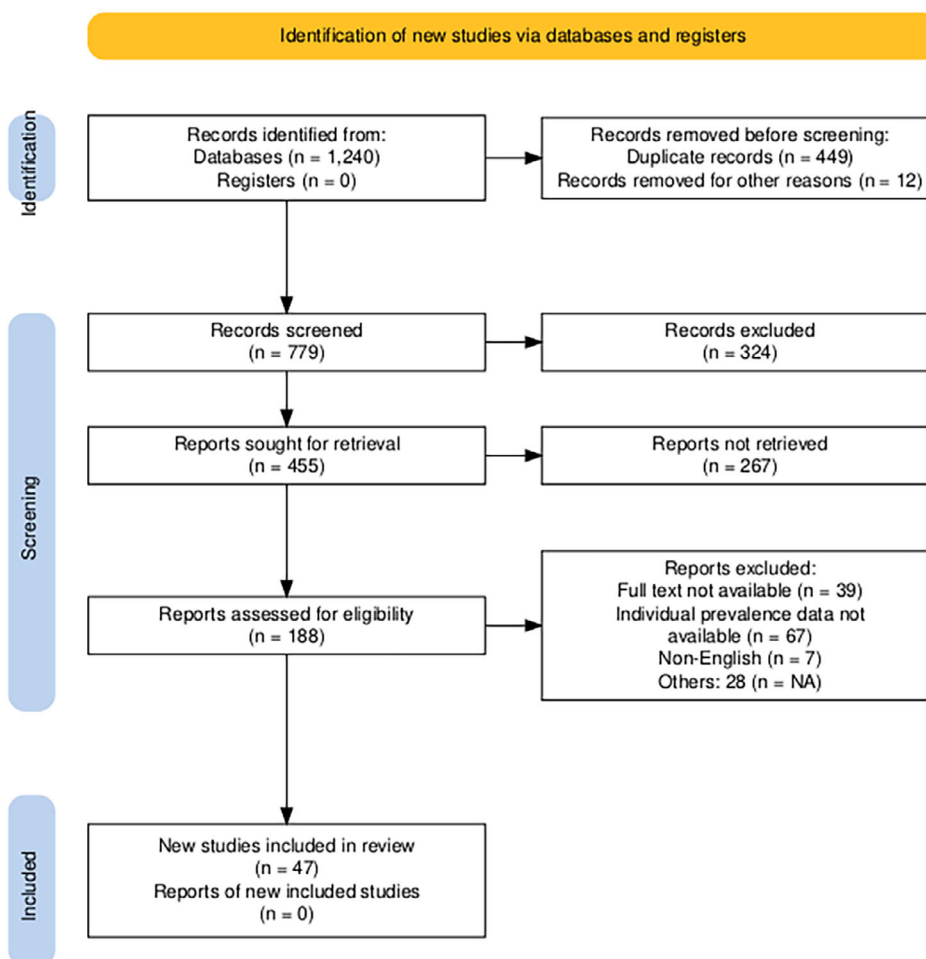


FIGURE 1 | PRISMA flow diagram for the inclusion of eligible articles in the meta-analysis.

In these studies, a total of 166,282 samples were analysed, and 19,188 individuals were identified as positive samples. Moreover, the highest and lowest sample sizes were 64,109 and 41, respectively. Prevalence rates varied significantly across regions. However, in one study, the prevalence of LSD was 76.88% (95% CI: 72.53–80.97%) in Pakistan (Khan et al. 2024), and the lowest rate was 4.51% (95% CI: 4.27–4.75%) in Turkey (Mat et al. 2021).

3.3 | Meta-Analysis of LSD in Asia

The estimated overall pooled prevalence of LSD was 25.22% (95% CI: 19.95–30.88%, τ^2 : 0.19, I^2 : 99.82, $p < 0.0001$) in cattle and

buffalo according to random effects meta-analysis. The forest plot illustrates the pooled prevalence of LSD in Asia (Figure 2). The individual study weights showed minimal variation from 1.92% to 2.16%.

3.4 | Subgroup Meta-Analysis

A subgroup meta-analysis was performed for the species, country, diagnosis, disease occurrence, and year (Figures 3–5, Tables 2, 3). The total estimated pooled prevalence of LSD by species was 26.15% (95% CI: 20.73–31.96%, $p < 0.0001$) in cattle and 7.66% (95% CI: 3.60–13.00%, $p = 0.11$) in buffalo. Among the included Asian

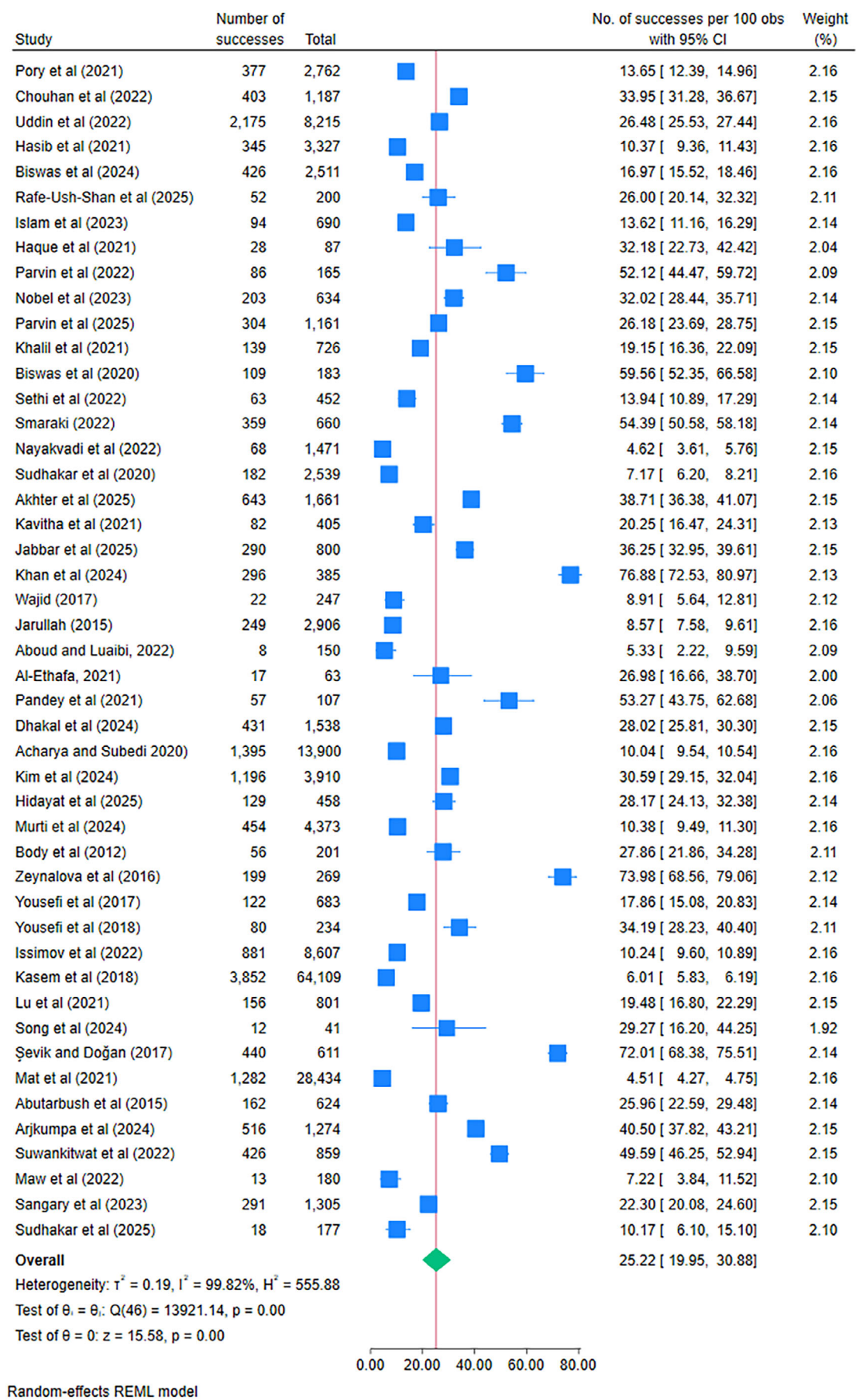


FIGURE 2 | Forest plot of the meta-analysis of LSD in Asia.

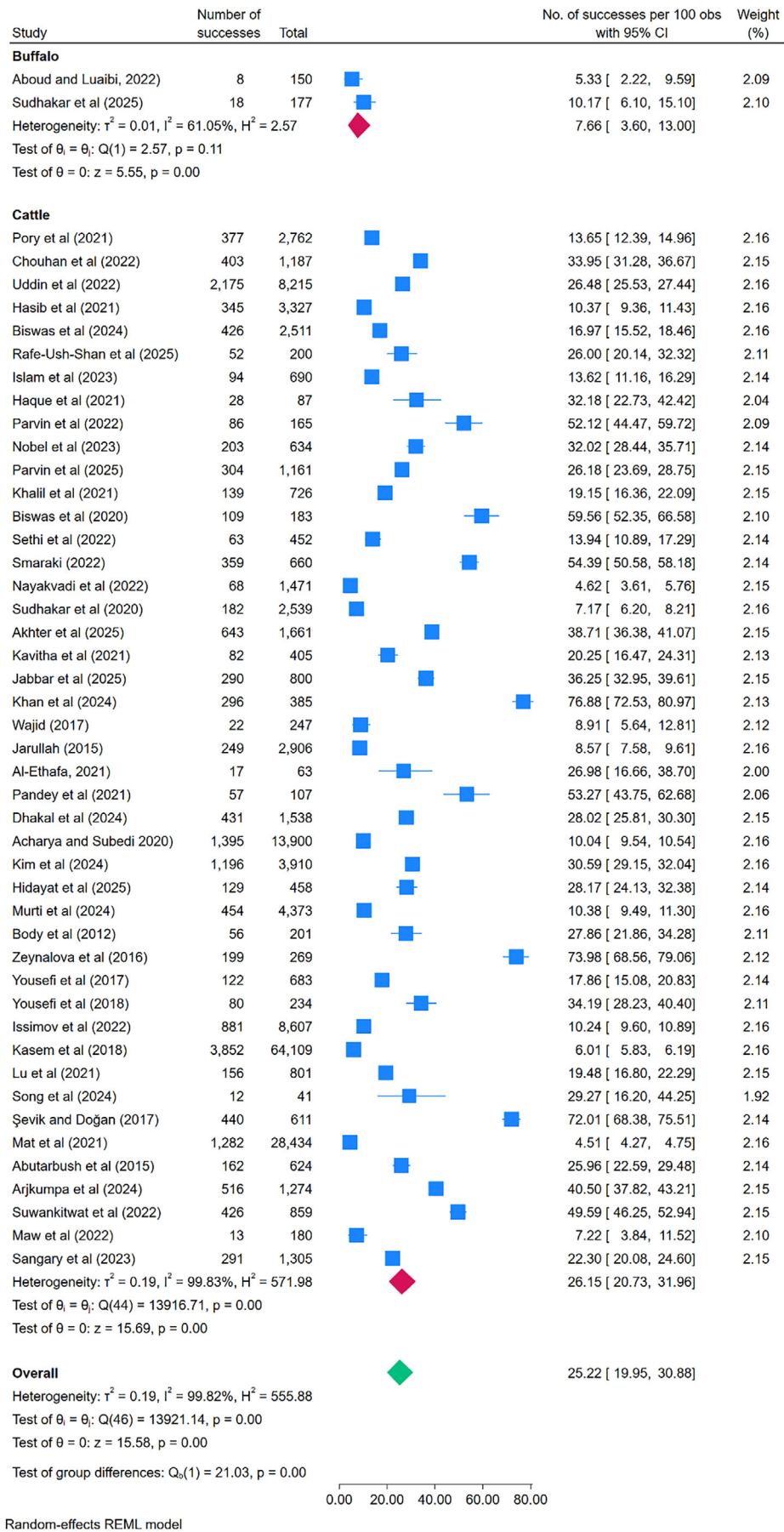


FIGURE 3 | Subgroup meta-analysis by species.

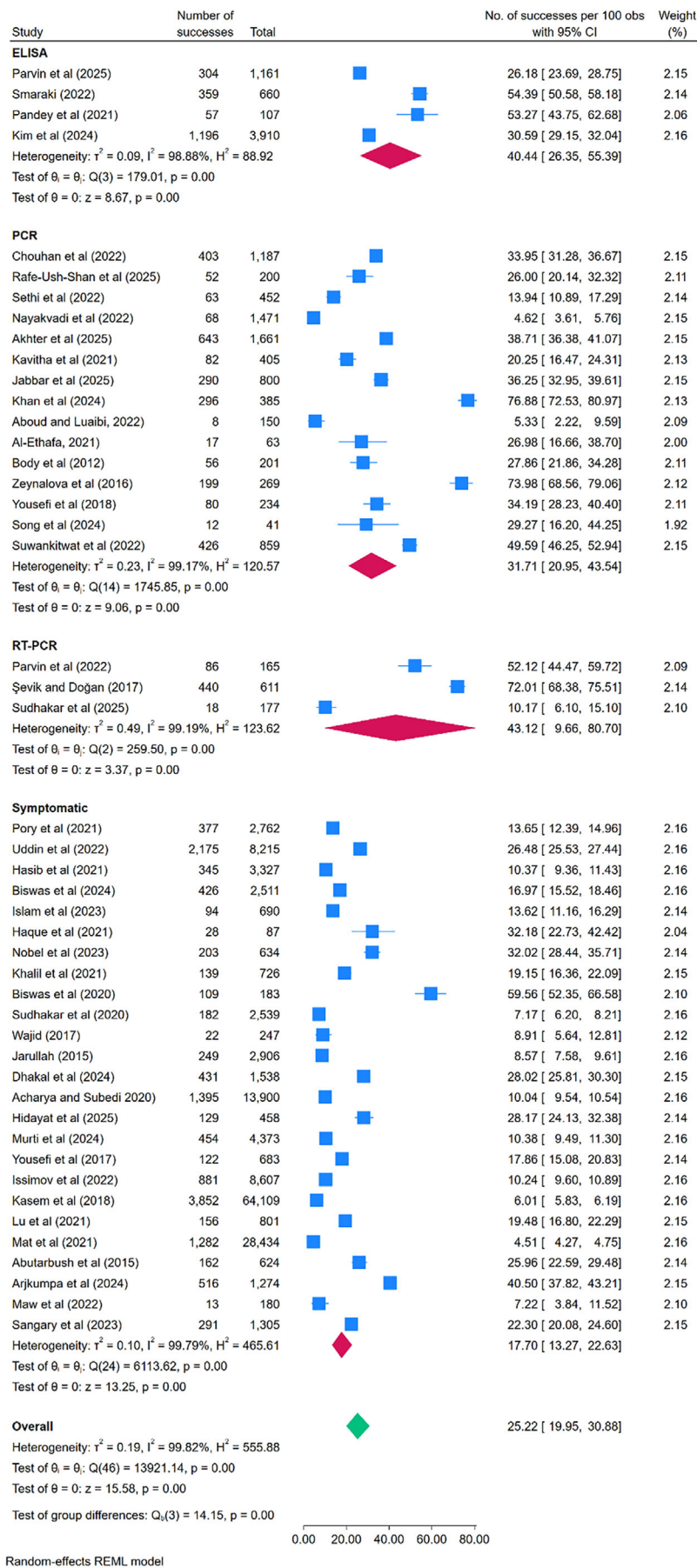


FIGURE 4 | Subgroup meta-analysis by diagnosis.

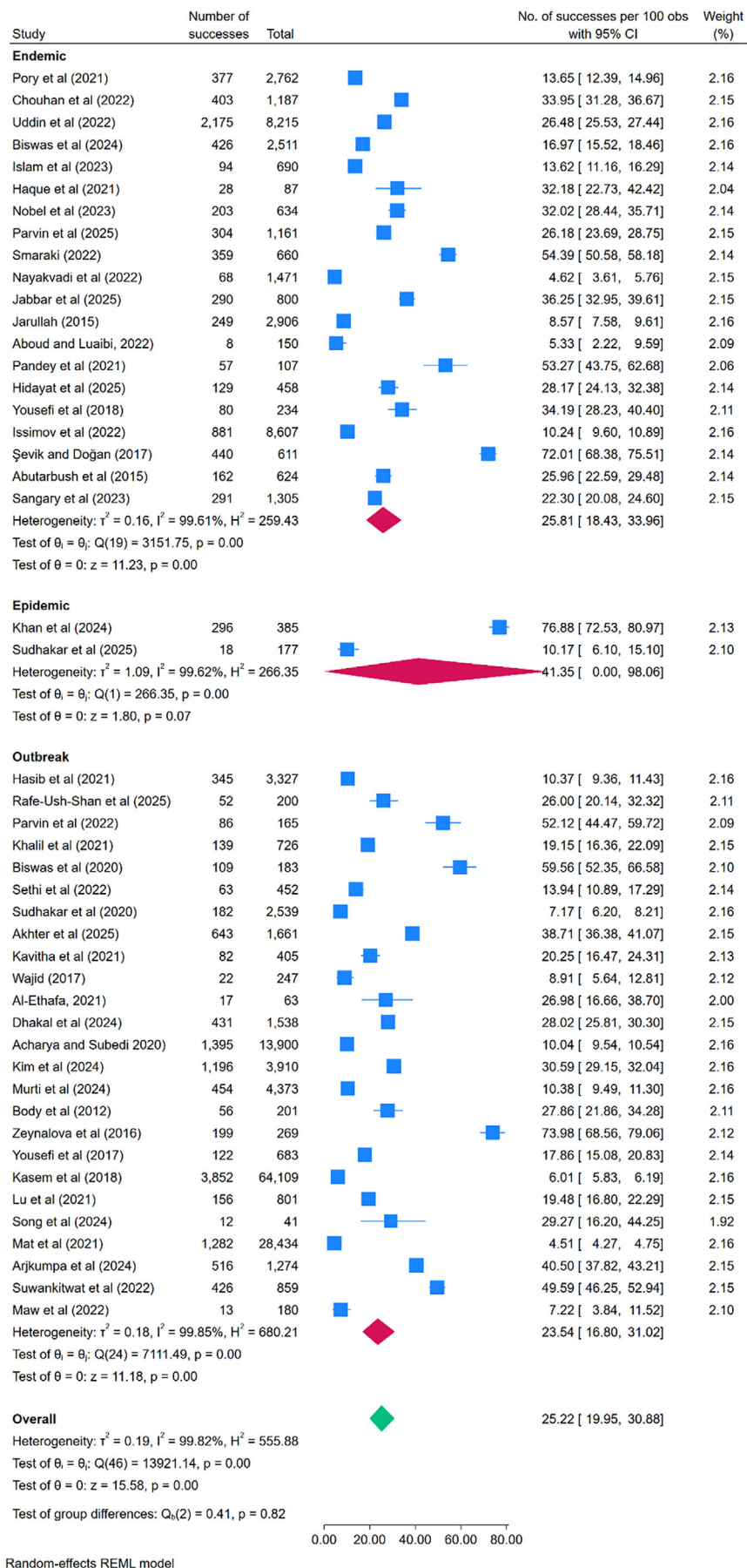


FIGURE 5 | Subgroup meta-analysis by disease occurrence.

TABLE 2 | Subgroup meta-analysis by country.

Country	Number of articles	Total	Positive (LSD)	Pooled prevalence	95% CI	τ^2	I^2 (%)	<i>p</i> -value	H^2
Afghanistan	1	1305	291	22.3	20.08–24.60	0.00	—*	—	—
Azerbaijan	1	269	199	73.98	68.56–79.06	0.00	—	—	—
Bangladesh	13	21,848	4741	26.65	19.25–34.77	0.10	99.3	< 0.0001	149.10
China	2	842	168	22.08	13.73–31.71	0.02	55.57	< 0.0001	2.25
India	7	7365	1415	19.10	7.95–33.61	0.20	99.47	< 0.0001	189.74
Indonesia	2	4831	583	18.34	4.54–38.56	0.11	98.88	< 0.0001	89.16
Iran	2	917	202	25.42	11.31–42.38	0.07	95.96	< 0.0001	24.75
Iraq	4	3366	296	10.86	4.36–19.70	0.06	93.98	< 0.0001	16.62
Jordan	1	624	162	25.96	22.59–229.48	0.00	—	—	—
Kazakhstan	1	8607	881	10.24	9.60–10.89	0.00	—	—	—
Myanmar	1	180	13	7.22	3.84–11.52	0.00	—	—	—
Nepal	3	15,545	1883	28.41	7.72–55.63	0.24	99.72	< 0.0001	355.81
Oman	1	201	56	27.86	21.86–34.28	0.00	—	—	—
Pakistan	2	1185	586	57.14	18.33–91.34	0.35	99.46	< 0.0001	185.46
Saudi Arabia	1	64,109	3852	6.01	5.83–6.19	0.00	—	—	—
South Korea	1	3910	1196	30.59	29.15–32.04	0.00	—	—	—
Thailand	2	2133	942	44.97	36.18–53.93	0.02	94.17	< 0.0001	17.16
Turkey	2	29,045	1722	33.11	0.00–97.02	1.28	99.93	< 0.0001	1528.42

*Not calculated because only one investigation was available on this topic.

TABLE 3 | Subgroup meta-analysis by year.

Year	Total	Positive (LSD)	Pooled prevalence	95% CI	τ^2	I^2 (%)	<i>p</i> -value	H^2
2012	201	56	27.86	21.86–34.28	0.00	—*	—	—
2015	3530	411	16.26	3.25–36.47	0.11	99.14	< 0.0001	116.22
2016	269	199	73.98	68.56–79.06	0.00	—	—	—
2017	1541	584	30.51	2.43–71.88	0.56	99.63	< 0.0001	270.85
2018	64,343	3932	17.69	0.32–51.97	0.28	99.25	< 0.0001	133.23
2020	16,622	1686	22.11	1.09–58.58	0.45	99.90	< 0.0001	1043.36
2021	37,986	2999	22.23	13.79–32.00	0.12	99.54	< 0.0001	218.20
2022	21,946	4482	23.13	11.74–36.98	0.24	99.75	< 0.0001	401.83
2023	2629	588	22.18	12.71–33.39	0.05	97.53	< 0.0001	40.52
2024	12,935	2833	27.45	12.46–45.62	0.26	99.76	< 0.0001	410.32
2025	4280	1418	31.21	26.03–36.64	0.02	92.22	< 0.0001	555.88

*Not calculated because only one investigation was available on this topic.

countries, Azerbaijan presented the highest pooled prevalence (73.98%, 95% CI: 68.56–79.06%), whereas Saudi Arabia reported the lowest estimate (6.01%) (Table 2 and Figure 6). Disease occurrence categories revealed that epidemic cases had the highest prevalence (41.35%, 95% CI: 0–98.06%, *p* < 0.0001), followed by endemic cases (25.81%, 95% CI: 18.43–33.96%, *p* = 0) and outbreak

cases (23.54%, 95% CI: 16.80–31.02%, *p* < 0.0001). Diagnostic modality significantly influenced the prevalence estimates, with RT-PCR (43.12%, 95% CI: 9.66–80.70%, *p* < 0.0001) and ELISA (40.44%, 95% CI: 26.35–55.39%, *p* < 0.0001) detecting higher case rates than PCR (31.71%, 95% CI: 20.95–43.54%, *p* < 0.0001) for clinical signs diagnosis (17.70%, 95% CI: 13.27–22.63%, *p* < 0.0001).

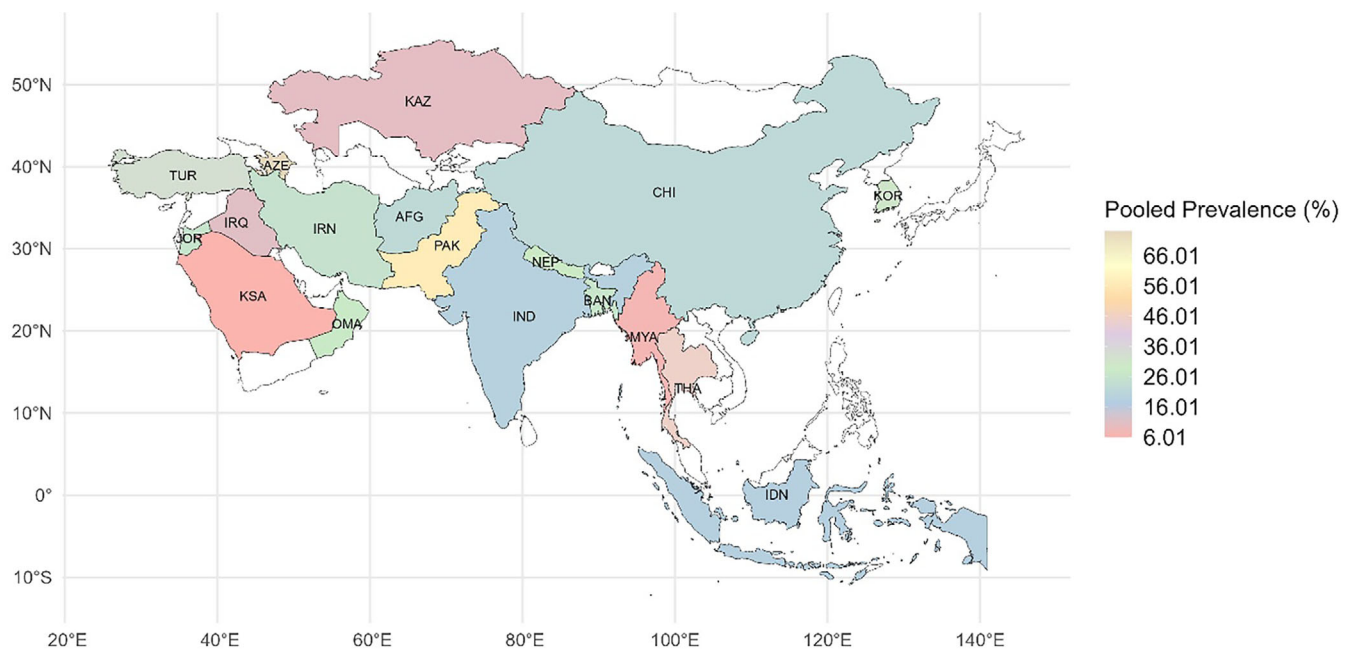


FIGURE 6 | Countries with pooled prevalence rates of LSD in different Asian countries.

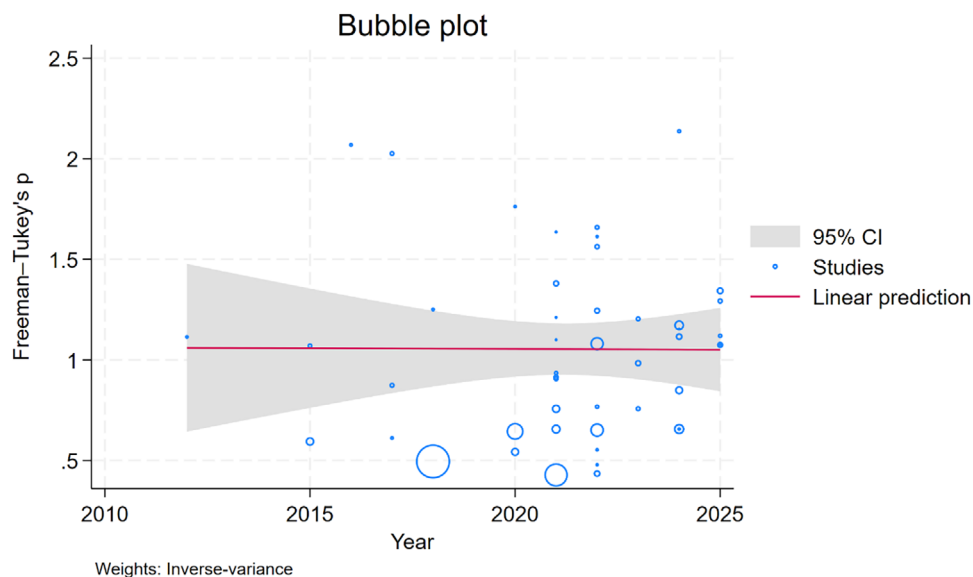


FIGURE 7 | Meta-regression of the prevalence of LSD in cattle and buffalo and year of study. Each circle represents an individual study, with the size of the circle proportional to the study's inverse-variance weight, reflecting the relative influence of each study on the overall analysis.

Temporal analysis revealed a peak prevalence in 2016 (73.98%, 95% CI: 68.56–79.06%) and the lowest estimate in 2015 (16.26%, 95% CI: 3.25–36.47%, $p < 0.0001$) (Table 3).

the linear prediction line and 95% confidence intervals indicating substantial variability. The residual I^2 remained high at 99.79%, suggesting that the year accounts for minimal heterogeneity.

3.5 | Meta-Regression

To explore the sources of heterogeneity in our meta-analysis of the prevalence of LSD in cattle and buffalo, we conducted a meta-regression analysis according to the year of study. The bubble plot illustrates the relationship between prevalence and publication year (Figure 7). A slight, non-significant downward trend was observed ($\beta = -0.0006$, 95% CI: -0.0437 to 0.0423 , $p = 0.975$), with

3.6 | Publication Bias

To assess potential publication bias in our meta-analysis of the prevalence of LSD in Asia, we conducted a funnel plot analysis and Egger's test. The funnel plot revealed an asymmetrical distribution of effect sizes, suggesting publication bias (Figure 8). Egger's test, which uses a random-effects model (REML), regressed the Freeman–Tukey transformed prevalence

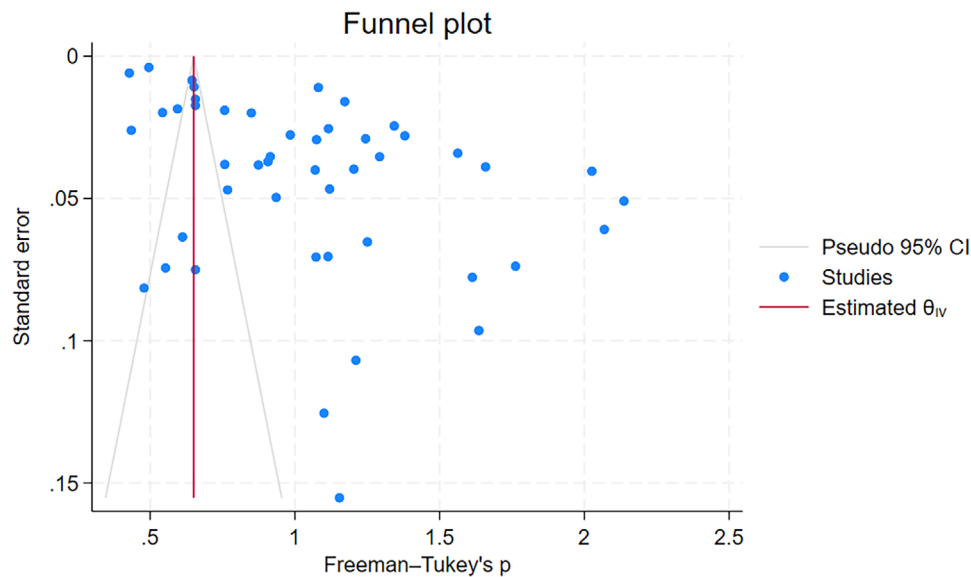


FIGURE 8 | A funnel plot illustrating publication bias in different studies.

against its standard error, yielding an intercept (β_0) of 4.01 (SE = 1.993, $z = 2.01$, $p = 0.0442$), indicating significant small-study effects ($p < 0.05$) and potential publication bias. These findings support the robustness of our meta-analysis results.

3.7 | Sensitivity Analysis

To evaluate the robustness of our meta-analysis results on disease incidence, we conducted a leave-one-out sensitivity analysis via the random-effects REML model. Each of the 47 studies was systematically excluded one at a time, and the pooled prevalence was recalculated to assess the influence of individual studies on the overall estimate. The forest plot (Figure 9) illustrates the results, with the pooled prevalence ranging from 24% (95% CI: 19–30%) to 26% (95% CI: 21–32%) across all iterations, compared with the overall estimate of 25% (95% CI: 20–31%). Notably, excluding ‘Khan et al. (2024)’ resulted in the lowest pooled prevalence of 24% (95% CI: 19–29%, $p < 0.000$), whereas omitting ‘Nayakvadi et al. (2022)’ yielded the highest at 26% (95% CI: 21–32%, $p < 0.000$). The consistency of the pooled estimate across exclusions, with all p -values remaining significant ($p < 0.000$), indicates that no single study disproportionately influenced the overall prevalence. This stability enhances confidence in the reliability of our findings, despite the high heterogeneity observed ($I^2 > 90\%$).

4 | Discussion

To the author’s knowledge, this is the first systematic review and meta-analysis on LSD prevalence in Asia only. This systematic review and meta-analysis illustrated the results from 47 articles, highlighting the prevalence of LSD between 2012 and 2025 in different Asian countries. This study provides reliable prevalence estimates, contributing to a deeper understanding of LSD epidemiology. These insights are vital for enhancing efforts in the control and eradication of LSD.

Among the articles, 13 were from Bangladesh, seven from India, two from Pakistan, four from Iraq, three from Nepal, one from South Korea, two from Indonesia, one from Oman, one from Azerbaijan, two from Iran, one from Kazakhstan, one from Saudi Arabia, two from China, two from Turkey, one from Jordan, two from Thailand, one from Myanmar, and one from Afghanistan. A relatively large number of studies were from Bangladesh.

The analysis revealed an estimated pooled prevalence of 25.22% (95% CI: 19.95–30.88%) across 47 published studies. A study by Abebaw (2024) reported a high LSD prevalence of 54% in Africa, with transmission occurring mainly through blood-feeding insects such as flies, mosquitoes, and ticks. These vectors are abundant in many African countries, particularly in hot and humid regions where the disease is prevalent. For example, studies have shown that ixodid (hard) ticks can transmit LSDVs through transstadial and transovarial transmission, as well as mechanical or interstadial transmission (Tuppurainen et al. 2011). The abundance of these vectors in Africa creates an environment conducive to accelerating disease transmission. This study revealed significant differences in LSD prevalence in Asian countries.

Vector ecology, climatic conditions, animal movement, and viral evolution influence LSDV transmission in Asia. Hematophagous arthropods such as ticks and mosquitoes play crucial roles in the spread of LSDV, with ticks exhibiting overwintering capabilities that sustain the virus in colder months (Khalafalla 2022). Warm, humid climates and seasonal rainfall further promote vector proliferation, increasing LSDV outbreaks (Agrawal et al. 2024). Cross-border and local animal movements significantly spread the disease, particularly in countries such as Vietnam and Thailand (Sprygin et al. 2022; Le et al. 2024). Additionally, the emergence of recombinant strains, including vaccine escape variants, has complicated control efforts (Wang et al. 2022). Limited awareness, vaccination gaps, and cultural practices such as communal grazing exacerbate disease transmission (Le et al. 2024). This study highlights the need for several studies on LSD in Asia and other continents to understand the disease

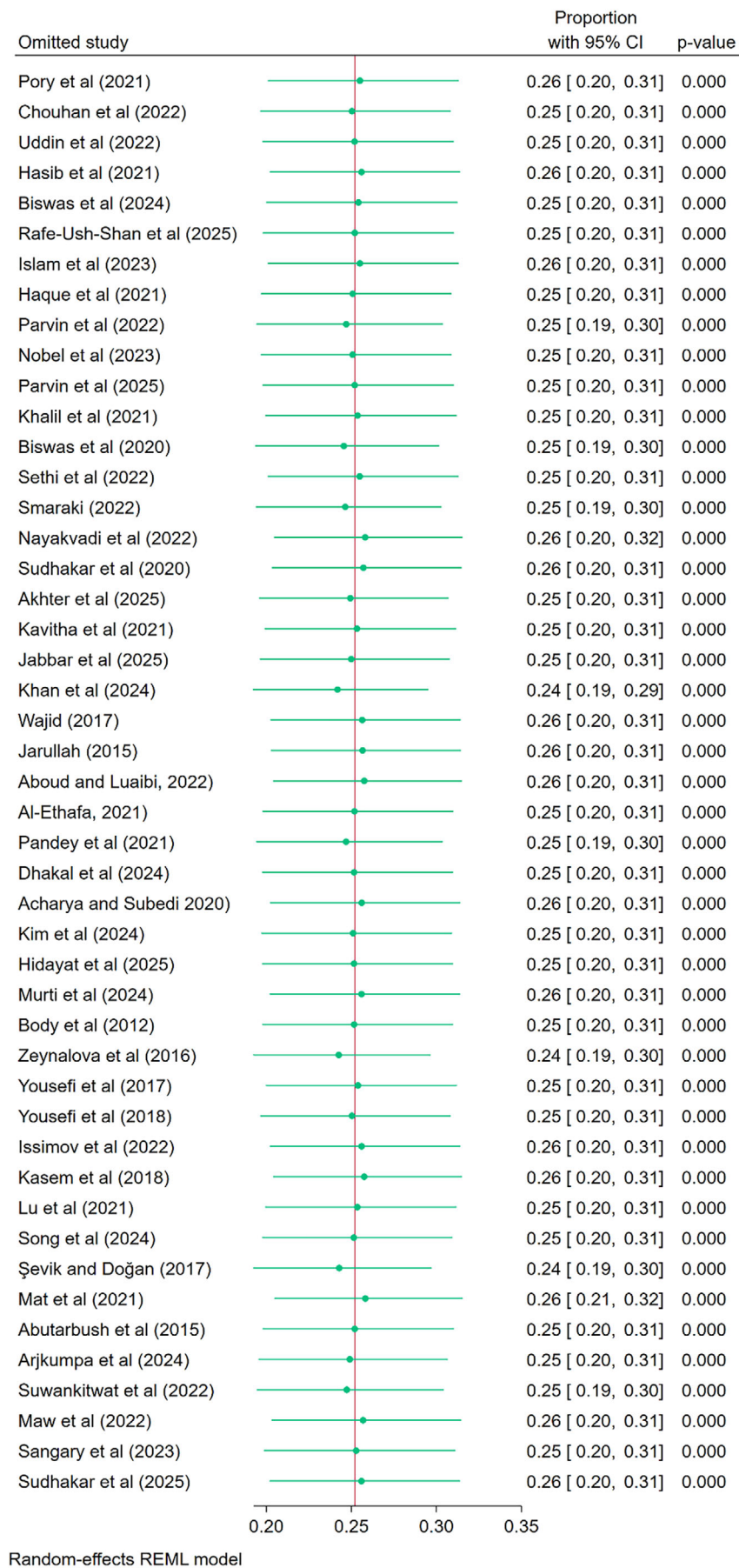


FIGURE 9 | Sensitivity analysis of the prevalence of LSD in Asian countries from 2012 to 2025.

properly for effective control of this economically significant disease.

Our meta-analysis revealed a regional variation in the pooled prevalence of LSD across Asian countries, highlighting the heterogeneous disease burden in the region. The highest prevalence was observed in Azerbaijan (73.98%), followed by Pakistan (57.14%), Thailand (44.97%), and Turkey (33.11%), whereas Saudi Arabia (6.01%), Myanmar (7.22%), and Kazakhstan (10.24%) reported the lowest rates. This may be due to differences in climate, vector distribution, and livestock management practices that contribute to regional disparities in LSD prevalence across Asia (Lubinga et al. 2014; Sprygin et al. 2022; Li et al. 2023). In South Asia, Pakistan's notably high prevalence (57.14%) contrasts with that of India (19.10%) and Bangladesh (26.65%), potentially reflecting differences in vaccination coverage and disease surveillance (Tuppurainen and Oura 2012; Sprygin et al. 2022). Similarly, the Middle East showed diverse estimates, with Saudi Arabia's low prevalence (6.01%), possibly due to stringent control measures, whereas Oman (27.86%) and Jordan (25.96%) reported higher rates, likely influenced by cross-border cattle movement (Abutarbush et al. 2015). In East Asia and Southeast Asia, Thailand's high prevalence (44.97%) compared with Indonesia (18.34%) and China (22.08%) may be linked to tropical climates favouring vector proliferation, a factor (Afshari Safavi 2022).

Compared with PCR (31.71%), ELISA (40.44%), and clinical signs diagnosis (17.70%), RT-PCR (43.12%) demonstrated a greater prevalence of LSD because of its superior sensitivity in detecting viral DNA and identifying more active infections, including subclinical cases (Jiang et al. 2022). Unlike conventional PCR, which may miss low viral loads, RT-PCR's real-time fluorescence detection enhances its ability to identify early infections, whereas ELISA, which targets antibodies, reflects exposure rather than active disease, and clinical signs diagnosis overlooks cases without clinical signs (Stubbs et al. 2012).

The findings of this review suggest a greater demand for LSD research in Asia. The economic losses caused by LSD, including reduced milk production and hide value, create a disincentive for farmers to invest in prevention measures. In Pakistan, for example, LSD has led to a significant decline in milk and meat production (Haider et al. 2023). Milk production can be reduced by more than 50% in infected herds and those infected with secondary infections (Selim et al. 2021). LSD causes severe economic losses in dairy and beef production across Asia, with meat consumption decreasing by 60% to 70% in regions such as Pakistan (Di Giuseppe et al. 2024). The disease also decreases hide value due to skin lesions, while morbidity rates can reach 100%, leading to secondary complications and reproductive losses in affected herds (Haider et al. 2023). Indirect losses stem from high vaccination costs, trade restrictions, and veterinary expenses, which disproportionately affect smallholder farmers (Le et al. 2024). Regional case studies highlight the financial burden, with India reporting a 30% to 55% drop in milk yield, Vietnam witnessing LSD spread to 55 provinces, and Thailand documenting average farm losses of \$727.38 per outbreak (Pandey et al. 2022; Le et al. 2024; Modethed et al. 2025).

Several limitations exist in this study, including the limited number of research articles from different countries involved and the

limited number of studies on buffaloes. Another limitation is that restricting the review to English-language articles may have led to the exclusion of pertinent studies published in other languages, potentially introducing language bias. This limitation could affect the comprehensiveness of the findings on the prevalence of LSD in Asia.

The substantial pooled prevalence of LSD in Asia, estimated at 25.2% overall with 26.2% in cattle and 7.7% in buffalo, indicates the necessity for a reassessment of regional disease control strategies. These results highlight the need for targeted surveillance systems that integrate molecular, clinical, and serological diagnostics to differentiate current infections from historical exposure accurately. The markedly high prevalence during epidemic years emphasises the importance of implementing mass vaccination programmes, with priority given to homologous live-attenuated vaccines due to their higher efficacy compared with heterologous alternatives, to rapidly achieve herd immunity. Coordinated animal movement restrictions, particularly during suspected outbreaks, combined with farmer education initiatives to enhance early disease reporting and farm-level biosecurity, are critical to limiting transmission and reducing economic losses. Furthermore, immune profiling through serological testing should be employed to guide the strategic allocation of vaccines and to support post-outbreak monitoring, thereby strengthening livestock health systems across diverse Asian regions.

5 | Conclusion

Bangladesh had the highest number of reports in this study, and the number of research articles in Asian countries was limited. Control of LSD outbreaks and increases in cattle and buffalo production in high-prevalence regions depend on regular screening tests and efficient preventative actions, including proper vaccination, the introduction of new animals, and vector control. In areas with viral circulation but insufficient prevalence data, epidemiological surveillance is crucial for accurately evaluating disease status and implementing targeted eradication plans. Accurate disease burden assessments depend on addressing elements causing variability in the estimation of prevalence.

Author Contributions

Md Jisan Ahmed: conceptualization, investigation, data extraction, data curation, formal analysis and interpretation of data, data visualization, writing – original draft, writing – review and editing, and project administration. **Ritu Chalise:** writing – original draft. **Md Imran Hossain:** writing – original draft. **Kazi Estieque Alam:** data visualization and data analysis. **Prajwal Bhandari:** data extraction and writing – original draft. **Md Abdur Rahman:** data extraction and writing – original draft. **Md Arifur Rahman:** data extraction. **Md Jayed Chowdhury:** data extraction. **Md Ismile Hossain Bhuiyan:** writing – original draft. **Delower Hossain:** data validation, data curation, writing – original draft, and writing – review and editing. **Mirza Synthia Sabrin:** writing – review and editing. **Mahabbat Ali:** writing – review and editing. All the authors read the full manuscript and agreed for publication.

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Ethics Statement

The authors have nothing to report.

Consent

The authors have nothing to report.

Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

Upon a reasonable request, the corresponding author can provide access to the datasets used and/or analysed in this study.

Peer Review

The peer review history for this article is available at <https://www.webofscience.com/api/gateway/wos/peer-review/10.1002/vms3.70607>.

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Supporting Information

Additional supporting information can be found online in the Supporting Information section.

Supplementary Materials S1: PRISMA 2009 Checklist.

Supplementary Materials S2 (Text): Quality index of the included articles. **Supplementary Materials S3:** Frequencies of quality categories of the selected studies. **Supplementary Materials S2 (Table):** Quality score and quality index score of individual contributing study.